

Section 'A'

- (1) Eqⁿ of circle passing through the origin centered on negative y-axis with radius a is $x^2 + y^2 + 2ay = 0$
- (2) The eqⁿ in cylindrical co-ordinate system of the equation in spherical coordinate system $\rho = 5 \cos \phi$ is $x^2 + z^2 = 5z$
- (3) If $\vec{u}$ is a differentiable vector function of t of constant length, then $\vec{u} \cdot \frac{d\vec{u}}{dt} = 0$
- (4) The curvature of a circle of radius a is $\frac{1}{a}$
- (5) A region in the plane is bounded if $\boxed{\text{'bounded' inside a finite circle}}$
- (6) The gradient of a differential function at a point is always $\boxed{\text{normal/perpendicular to level surface}}$
- (7) If $\vec{F}$ is conservative then the work done around any closed loop equals $\boxed{0}$
- (8) The area of the surface $f(x, y, z) = c$ over a closed and bounded plane region R is $A = \iint_R \frac{\sqrt{f_x^2 + f_y^2 + f_z^2}}{|f_z|} dx dy$
- (9) The Fourier coefficient a_n of the periodic function $f(t)$ of period T is given by the Euler's formula $\boxed{\frac{2}{T} \int f(t) \cos(2\pi n t / T) dt}$

(10) $\int_1^e \int_1^e \int_1^e \frac{1}{xyz} dx dy dz = \boxed{1}$

Section B

(11) The polar co-ordinates for Cartesian coordinate $(\sqrt{3}, 1)$ is $\boxed{2, \frac{\pi}{6}}$

(12) The eqn $\frac{8}{2-4\sin\theta}$ represents Hyperbola

(13) If $(x, y) = y \sin xy$ then $\frac{\partial f}{\partial y} = \boxed{\sin(xy) + xy \cos(xy)}$

(14) Unit tangent vector of a curve $\vec{r}(t) = \cos t \vec{i} + \sin t \vec{j}$ is $\boxed{\vec{T} = -\sin t \vec{i} + \cos t \vec{j}}$

(15) The ~~the~~ integral $\iint y S(x, y) dA$, where S is the density represents Moment

(16) The work done by the conservative field $\vec{F} = yz \vec{i} + xz \vec{j} + xy \vec{k}$ along any smooth curve C joining the point $(-1, 3, 9)$ to $(1, 6, -4)$ is $\boxed{3}$

(17) The divergence of $\vec{F} = 2xz \vec{i} - xy \vec{j} - \vec{k}$ is given by the expression $2z - x$

(18) $\int_0^\infty e^{-x^2} dx = \boxed{\frac{\sqrt{\pi}}{2}}$

19) The value of $\beta\left(\frac{1}{3}, \frac{2}{3}\right) = \frac{2\pi}{\sqrt{3}}$

20) $\frac{\pi}{2}$ is the primitive period of $f(x) = \sin 4x$.

Section - C

(2) Given, $f(x, y) = x^2 + xy$ at $P_0(1, 2)$

$$\vec{u} = \frac{1}{\sqrt{2}} \hat{i} + \frac{1}{\sqrt{2}} \hat{j}$$

Definition:

The directional derivative of f at P in direction $\vec{u}$ is

$$D_{\vec{u}} f(P) = \nabla f(P) \cdot \vec{u}$$

where $\vec{u}$ is a unit vector.

$$f_x = 2x + y$$

$$f_y = x$$

at $(1, 2)$

$$f_x = 4$$

$$f_y = 1$$

$$\therefore \nabla f(1, 2) = 4\hat{i} + \hat{j}$$

$$D_{\vec{u}} f = (4, 1) \cdot \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right) = \frac{4}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{5}{\sqrt{2}}$$

$$\text{For: } f(x, y) = \frac{x^2}{2} + \frac{y^2}{2}$$

$$\nabla f = (x, y)$$

The function increases most rapidly in the direction ∇f and decreases most rapidly in $-\nabla f$.

At a point (x, y) :

Max increase direction $= (x, y)$

Max decrease direction $= (-x, -y)$

(3)

The base is bounded by:

$$y = 4 - x^2$$

and $y = 3x$

Top surface: $z = x + 4$

$$4 - x^2 = 3x$$

$$x^2 + 3x - 4 = 0$$

$$(x+4)(x-1) = 0$$

$$x = -4, 1$$

$$-4 \leq x \leq 1$$

For each x

$$3x \leq y \leq 4 - x^2$$

$$\text{and } 0 \leq z \leq x + 4$$

For the volume:

$$V = \int_{-4}^1 \int_{3x}^{4-x^2} \int_0^{x+4} dz \, dy \, dx$$

$$V = \int_{-4}^1 (-x^3 - 7x^2 - 8x + 16) \, dx$$

$$V = \left[-\frac{x^4}{4} - \frac{7x^3}{3} - 4x^2 + 16x \right]_{-4}^1$$

$$\therefore V = \frac{625}{12}$$

4) Given

$$\vec{F} = (x^2 - y)\hat{i} + 4z\hat{j} + x^2\hat{k}$$

The curve is formed by the intersection of

$$\text{and } z = \sqrt{x^2 + y^2}$$

$$\therefore z = \sqrt{x^2 + y^2}$$
$$x^2 + y^2 = 4$$

So, C is a circle of radius 2

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 - y & 4z & x^2 \end{vmatrix}$$

$$= -4\hat{i} - 2x\hat{j} + \hat{k}$$

The surface is the disk $x^2 + y^2 \leq 4$, with upward normal

$$\nabla \times \vec{F} \cdot \hat{n} = 1$$

$$\therefore \oint_C \vec{F} \cdot d\vec{r} = \int_S 1 dA$$

Area of circle is:

$$\pi r^2 = \pi 2^2 = 4\pi$$

$$\therefore \oint_C \vec{F} \cdot d\vec{r} = 4\pi$$

(5) Given, $f(x, y) = \frac{x^4}{x^4 + y^2}$

$$\lim_{(x, y) \rightarrow (0, 0)} \frac{x^4}{x^4 + y^2}$$

(i) $y = 0, f(x, 0) = 1$

$$\therefore \lim_{x \rightarrow 0} f(x, 0) = 1$$

(ii) $x = 0, f(0, y) = 0$

$$\therefore \lim_{y \rightarrow 0} f(0, y) = 0$$

Since $1 \neq 0$

limits are different

$\therefore \lim_{(x, y) \rightarrow (0, 0)} \frac{x^4}{x^4 + y^2}$ does not exist

(6) Soln:

The Gamma function is:

$$\Gamma(n) = \int_0^{\infty} e^{-x} x^{n-1} dx$$

To prove $\Gamma(n+1) = n\Gamma(n)$

$$\Gamma(n+1) = \int_0^{\infty} e^{-x} x^{n+1-1} dx = \int_0^{\infty} e^{-x} x^n dx$$

$$= \left[x^n x e^{-x} / -1 \right]_0^{\infty} - \int_0^{\infty} nx^{n-1} \cdot \frac{e^{-x}}{-1} dx$$

$$= 0 + n \int_0^{\infty} e^{-x} x^{n-1} dx$$

$$= n\Gamma(n)$$

(7) Given, $r = \frac{e^\theta}{\sqrt{2}}$, $0 \leq \theta \leq \pi$

polar arc length: $L = \int_a^b \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$

(1) Find $dr/d\theta$ $\cdot \frac{dr}{d\theta} = \frac{e^\theta}{\sqrt{2}} = r$

$r^2 + \left(\frac{dr}{d\theta}\right)^2 = r^2 + r^2 = 2r^2$

$\sqrt{2r^2} = e^\theta$

$L = \int_0^\pi e^\theta d\theta$
 $= [e^\theta]_0^\pi$

$L = e^\pi - 1$

(8) Given,

$\frac{d\vec{r}}{dt} = (t^3 + 4t)\hat{i} + t\hat{j} + 2t^2\hat{k}$

$\vec{r}(0) = \hat{i} + \hat{j}$

Integrating we get,

$\vec{r}(t) = \left(\frac{t^4}{4} + 2t^2 + 1\right)\hat{i} + \left(\frac{t^2}{2}\right)\hat{j} + \frac{2t^3}{3}\hat{k}$

(9) Given, $\vec{r}(t) = \frac{t^3}{3} \hat{i} + \frac{t^2}{2} \hat{j}$, $t > 0$

$$\begin{aligned}\vec{r}' &= t^2 \hat{i} + t \hat{j} \\ &= t(t \hat{i} + \hat{j})\end{aligned}$$

$$|\vec{r}'| = \sqrt{t^4 + t^2} = t \sqrt{t^2 + 1}$$

$$T = \frac{\vec{r}'}{|\vec{r}'|} = \frac{t}{\sqrt{t^2 + 1}} \hat{i} + \frac{1}{\sqrt{t^2 + 1}} \hat{j}$$

Differentiate T:

$$T' = \frac{1}{(t^2 + 1)^{3/2}} \hat{i} - \frac{t}{(t^2 + 1)^{3/2}} \hat{j}$$

$$\therefore N = \frac{1}{\sqrt{t^2 + 1}} \hat{i} - \frac{t}{\sqrt{t^2 + 1}} \hat{j}$$

$$K = \frac{|\vec{r}' \times \vec{r}''|}{|\vec{r}'|^3}$$

$$= \frac{1}{t(t^2 + 1)^{3/2}}$$

(10) Given, $xy^2 = 54$
 $x = \frac{54}{y^2}$

$$\begin{aligned} \theta^2 &= x^2 + y^2 \\ &= \frac{2916}{y^4} + y^2 \end{aligned}$$

$$\frac{d\theta^2}{dy} = \frac{-11664}{y^5} + 2y$$

$$0 = \frac{-11664}{y^5} + 2y$$

$$\text{or, } 2y^6 = 11664$$

$$y^6 = 6^5$$

$$y^2 = 6^{\frac{5}{3}} = 6\sqrt[3]{6}$$

$$x = \frac{54}{y^2} = \frac{9}{3\sqrt[3]{6}}$$

So, the nearest points are: $\left(\frac{9}{3\sqrt[3]{6}}, \pm \sqrt{6\sqrt[3]{6}}\right)$

11) Soln: $I = \int_{-1}^1 \int_0^{2\pi} \int_0^{1+\cos\theta} 4r \, dr \, d\theta \, dz$

$$I = \int_{-1}^1 \int_0^{2\pi} \int_0^{1+\cos\theta} 2(1+\cos\theta)^2 \, d\theta \, dz$$

$$I = \int_{-1}^1 6\pi \, dz$$

$$\therefore I = 12\pi$$

(12) Given

$$\vec{F} = (x^2 + y)\hat{i} + (x + y^2)\hat{j}, \quad 0 \leq x \leq 1, \quad 0 \leq y \leq 1$$

$$\nabla \cdot \vec{F} = 2x + 2y \quad \text{Flux} = \int_0^1 \int_0^1 (2x + 2y) dx dy$$

$$= \int_0^1 (1 + 2y) dy$$

$$= 1 + 1$$

$$= 2$$

$\therefore$ Outward flux = 2

(13) Soln: $f(x) = \begin{cases} 0, & -\pi < x < 0 \\ x, & 0 < x < \pi \end{cases}$

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx = \pi/4$$

$$a_n = \frac{1}{\pi} \int_0^{\pi} x \cos nx dx = \frac{(-1)^n - 1}{\pi n^2}$$

$$b_n = \frac{1}{\pi} \int_0^{\pi} x \sin nx dx = \frac{(-1)^{n+1}}{\pi}$$

$$\therefore f(x) = \frac{\pi}{4} + \sum_{n=1}^{\infty} \frac{(-1)^n - 1}{\pi n^2} \cos nx + \frac{(-1)^{n+1}}{\pi} \sin nx$$